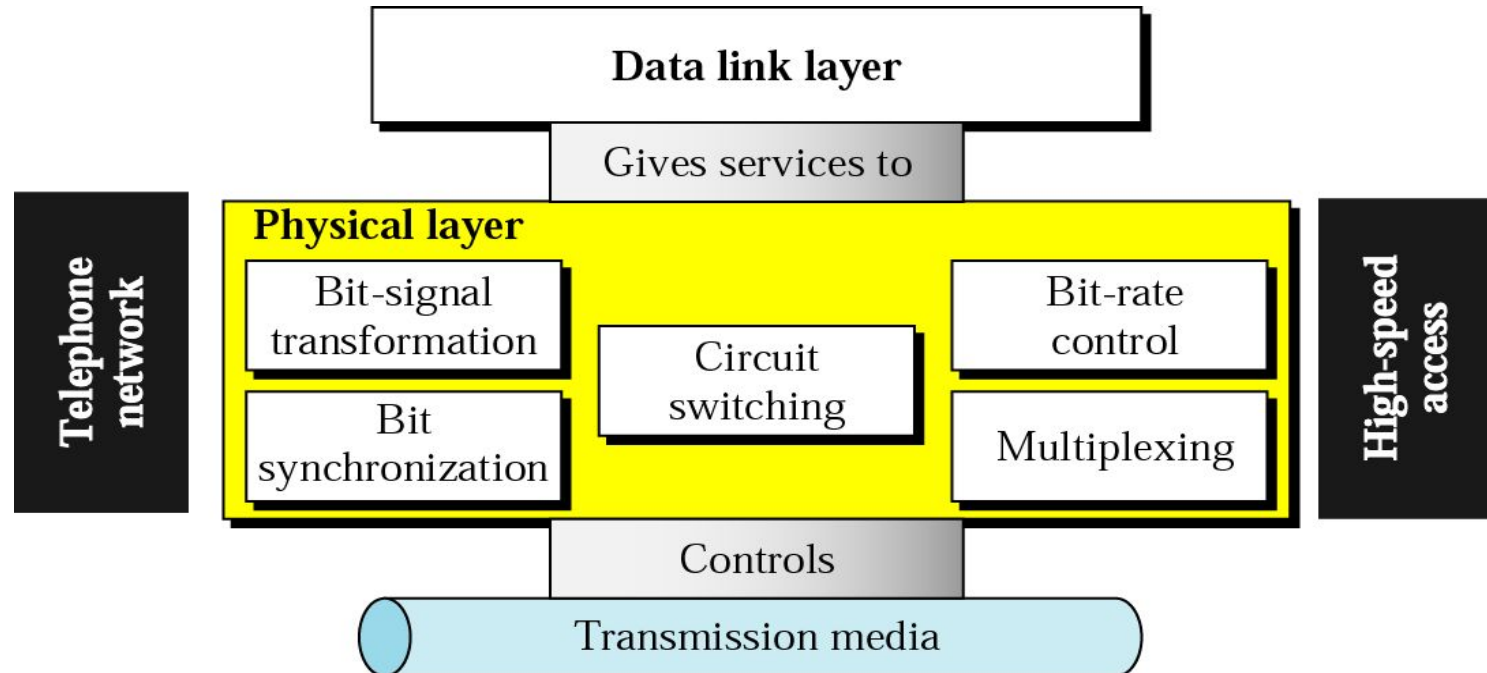


Computer Communication & Networks

Lecture 8

Physical Layer: Digital Transmission

Physical Layer



Physical Layer Topics to Cover

Signals

Digital Transmission

Analog Transmission

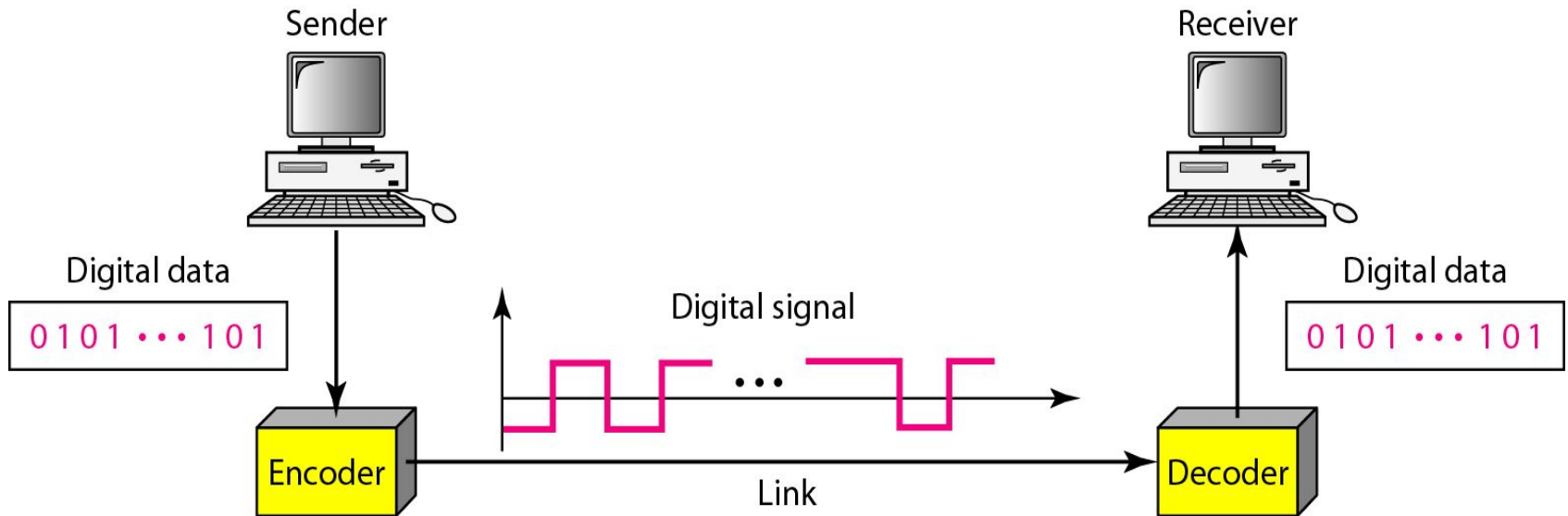
Multiplexing

Transmission Media

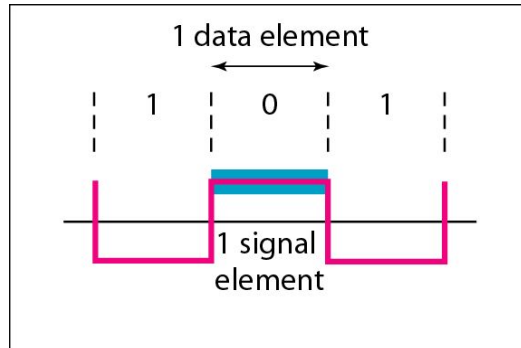
Digital to Digital Conversion

- *The conversion involves three techniques: **line coding**, **block coding**, and **scrambling**. Line coding is always needed; block coding and scrambling may or may not be needed.*
- *Line Coding*
- *Line Coding Schemes*
- *Block Coding*
- *Scrambling*

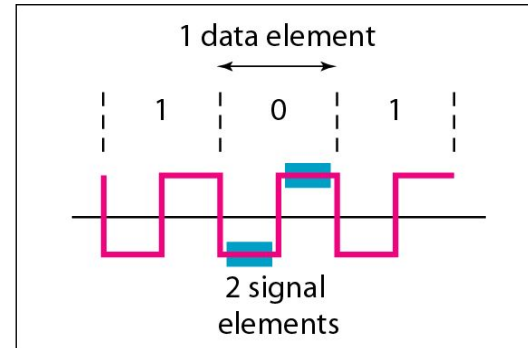
Line Coding & Decoding



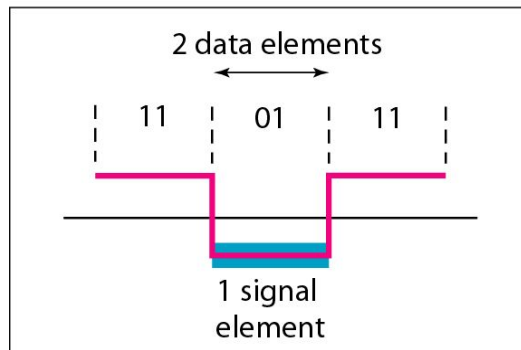
Signal Levels (Elements) Vs Data Levels (Elements)



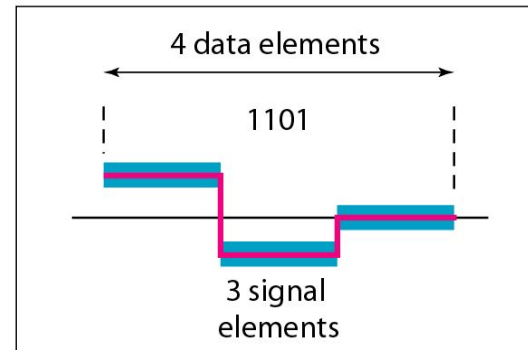
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Pulse Rate Vs Bit Rate

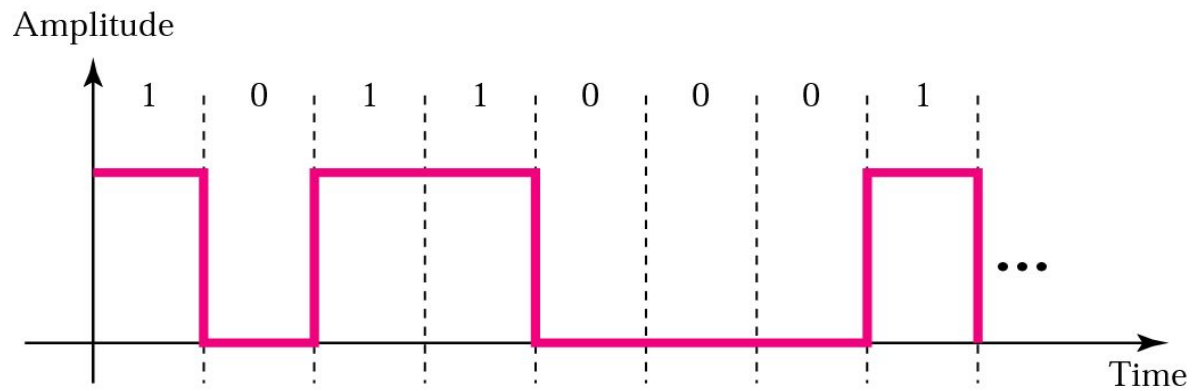
Example

A signal has two data levels with a pulse duration of 1 ms. We calculate the pulse rate and bit rate as follows:

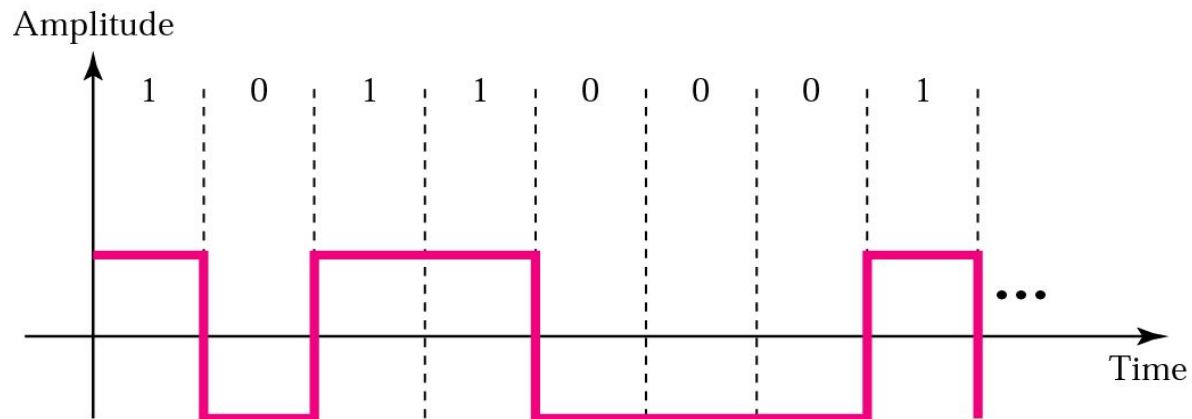
$$\text{Pulse Rate} = 1 / 10^{-3} = 1000 \text{ pulses/s}$$

$$\text{Bit Rate} = \text{Pulse Rate} \times \log_2 L = 1000 \times \log_2 2 = 1000 \text{ bps}$$

DC Component

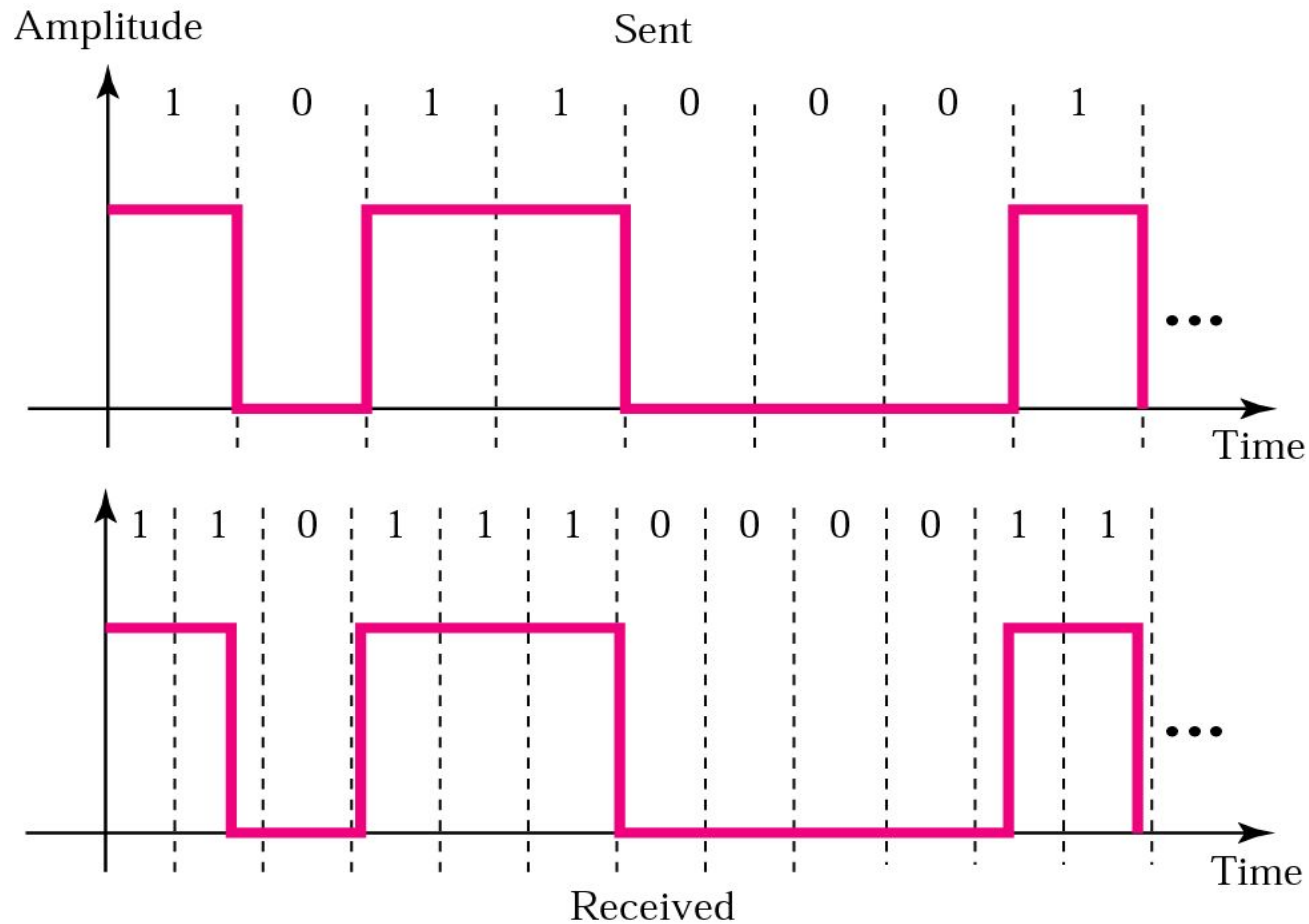


a. A signal with dc component



b. A signal without dc component

Lack of Synchronization



Example 3

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Kbps? How many if the data rate is 1 Mbps?

Solution

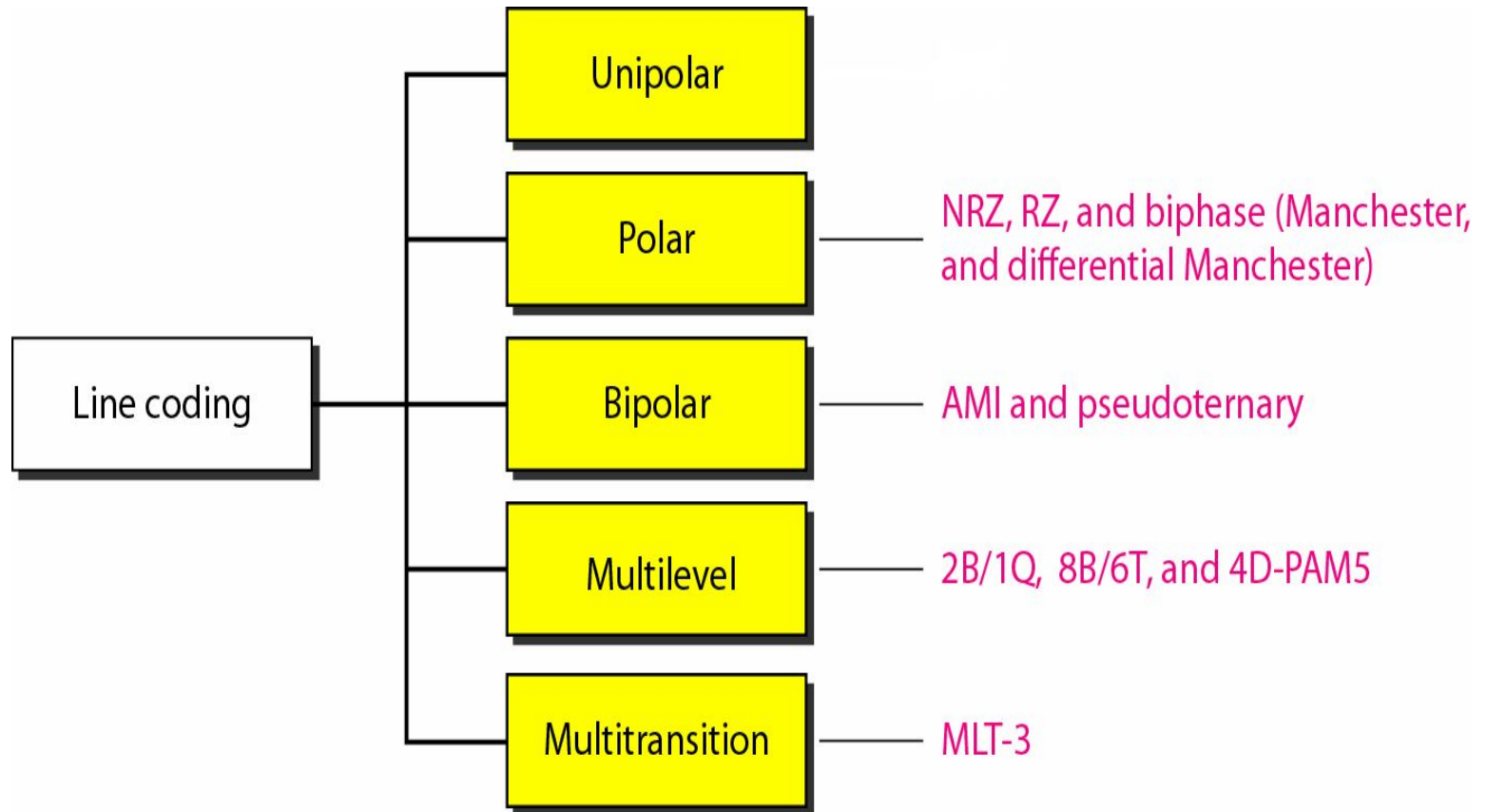
At 1 Kbps:

1000 bits sent $\square$ 1001 bits received $\square$ 1 extra bps

At 1 Mbps:

1,000,000 bits sent $\square$ 1,001,000 bits received $\square$ 1000 extra bps

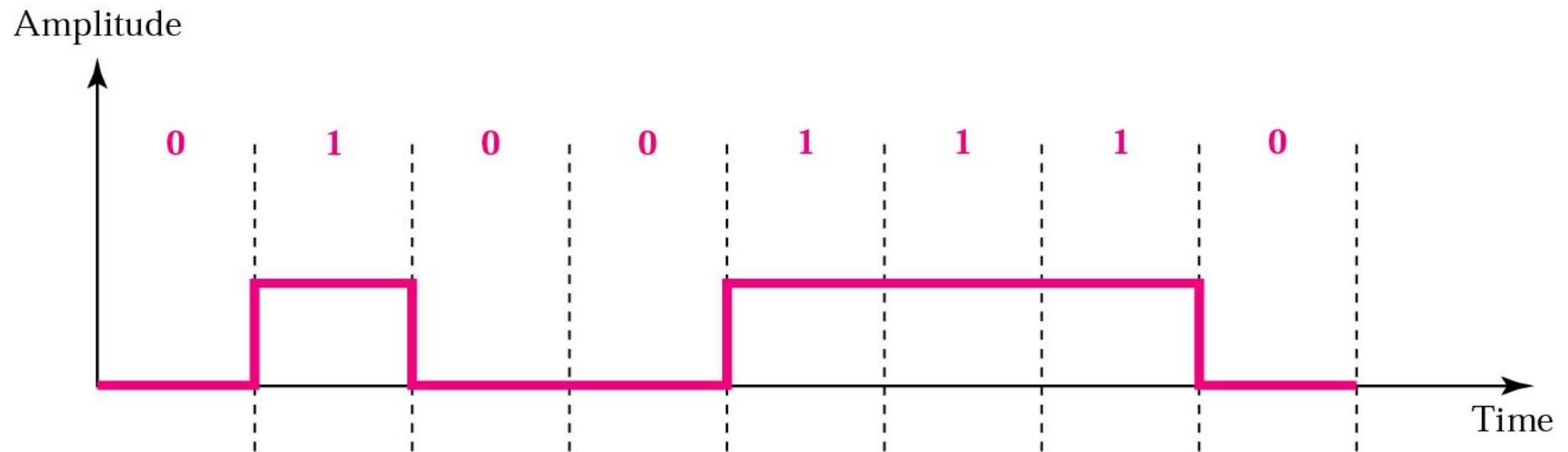
Line Coding Schemes



Note

In unipolar encoding, we use only one voltage level.

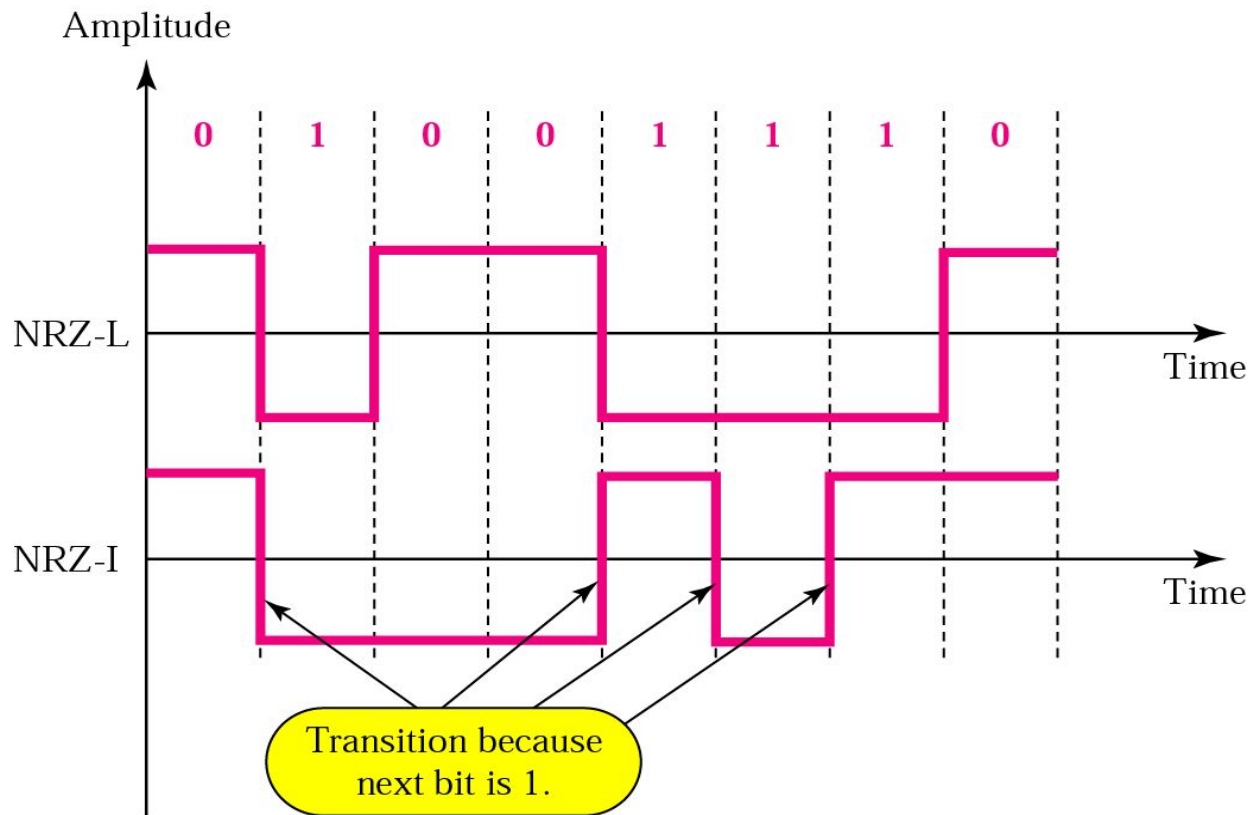
Unipolar Encoding



Note

In polar encoding, we use two voltage levels: positive & negative

Polar: NRZ-L and NRZ-I Encoding

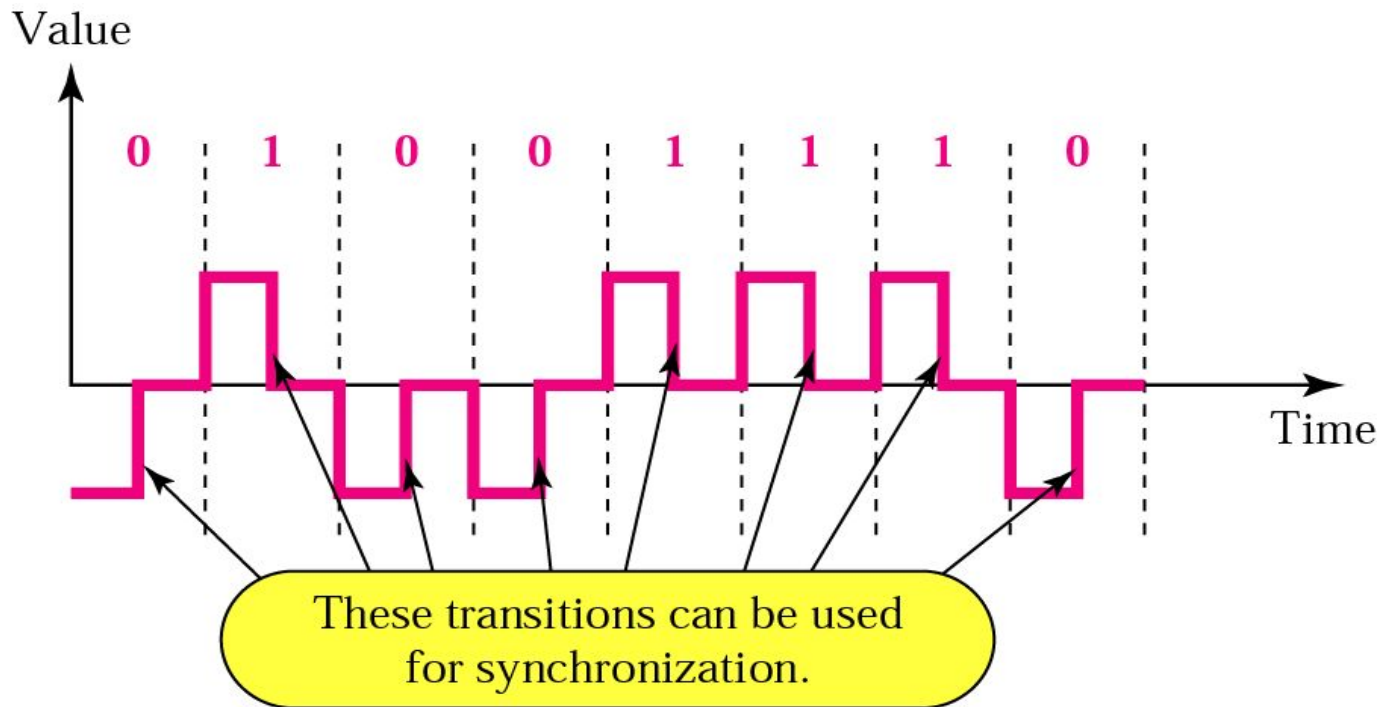


Note

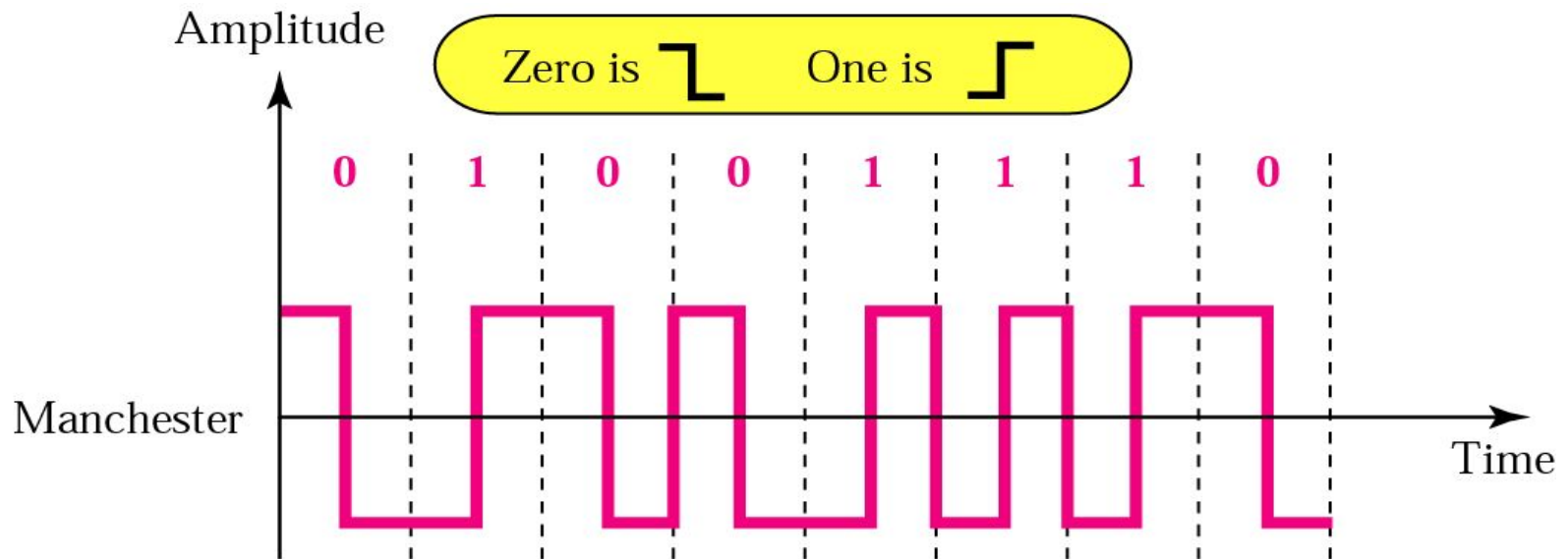
In NRZ-L the level of the voltage determines the value of the bit.

**In NRZ-I the inversion
or the lack of inversion
determines the value of the bit.**

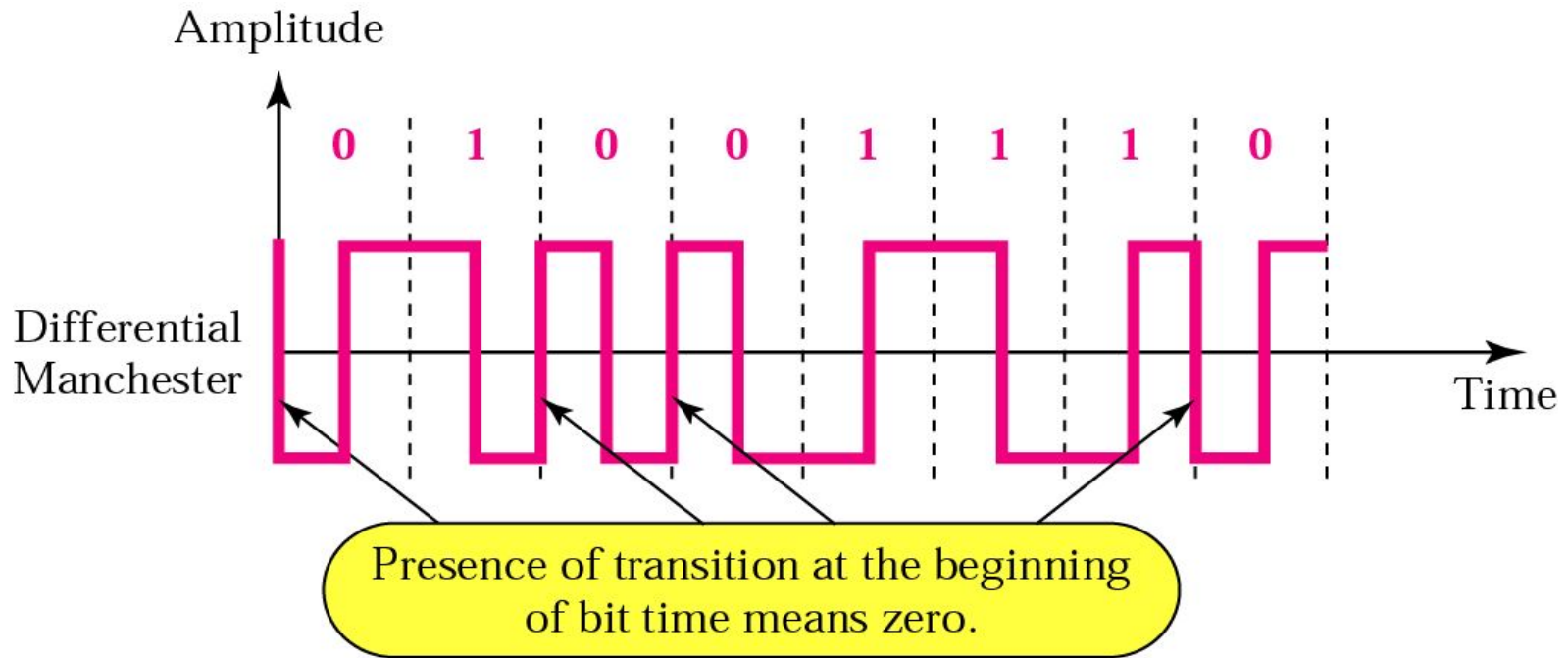
Polar: RZ Encoding



Polar: Manchester Encoding



Polar: Differential Manchester Encoding



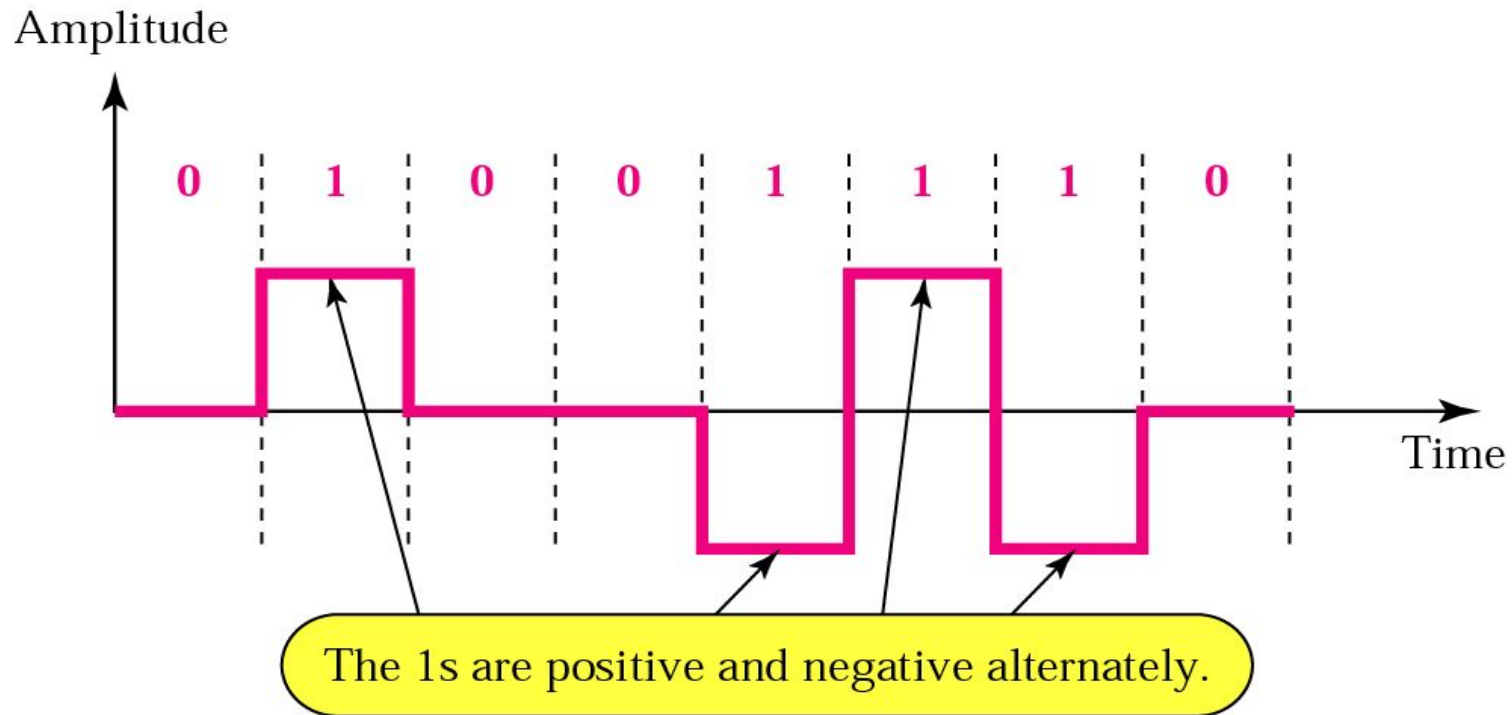
Note

In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

Note

**In bipolar encoding, we use three levels:
positive, zero, and negative.**

Bipolar: AMI (Alternative Mark Inversion) Encoding



Summary

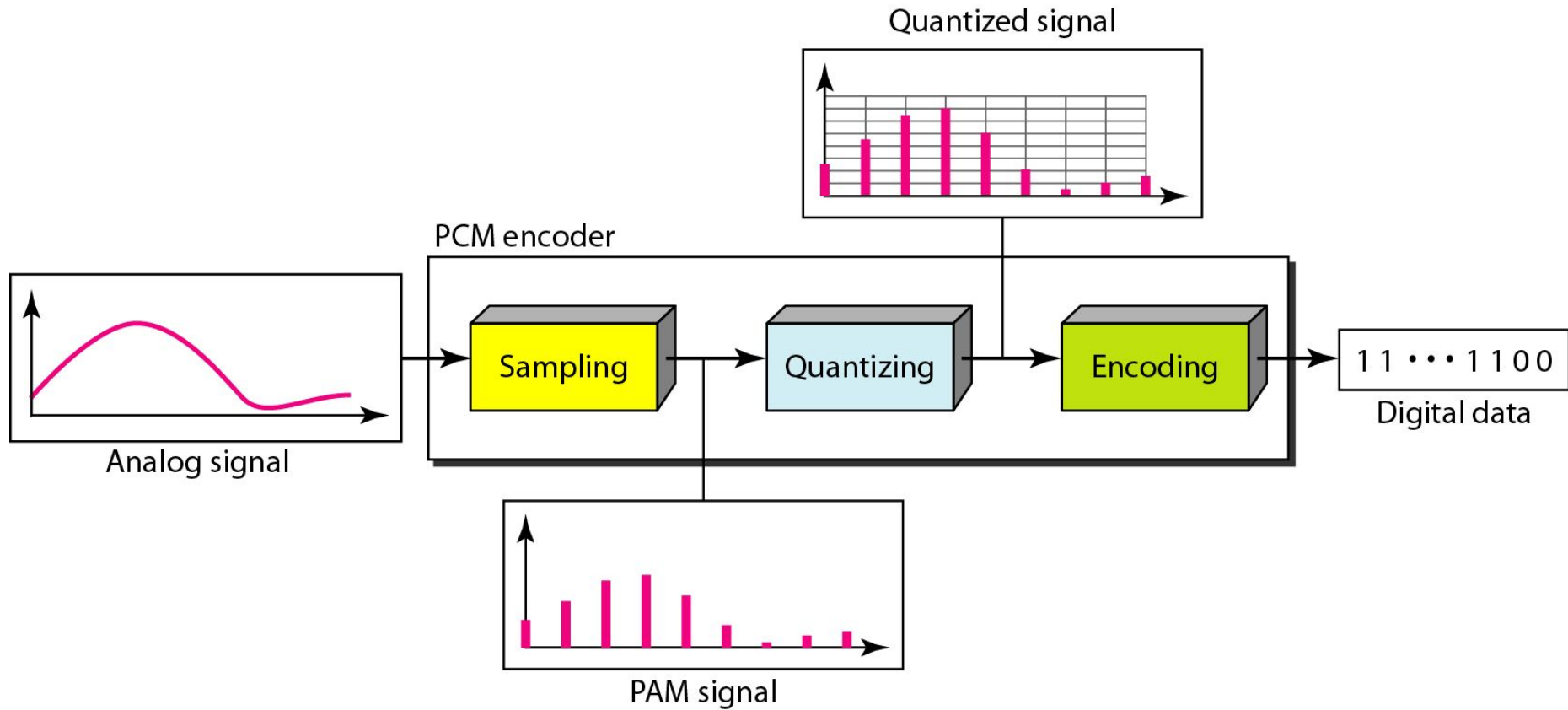
<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

Sampling

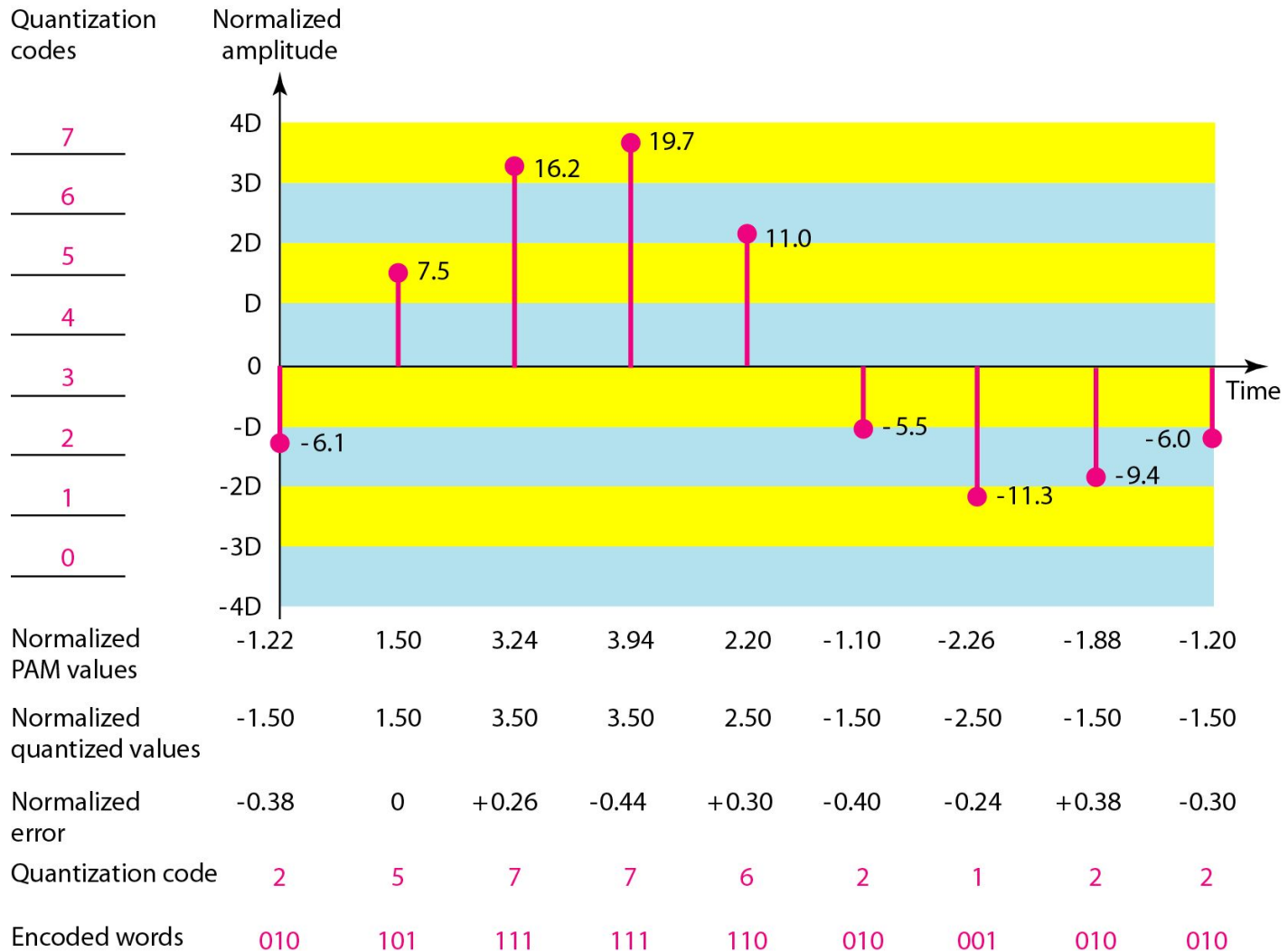
Pulse Code Modulation

Sampling Rate: Nyquist Theorem

PCM



Quantization & Encoding Samples



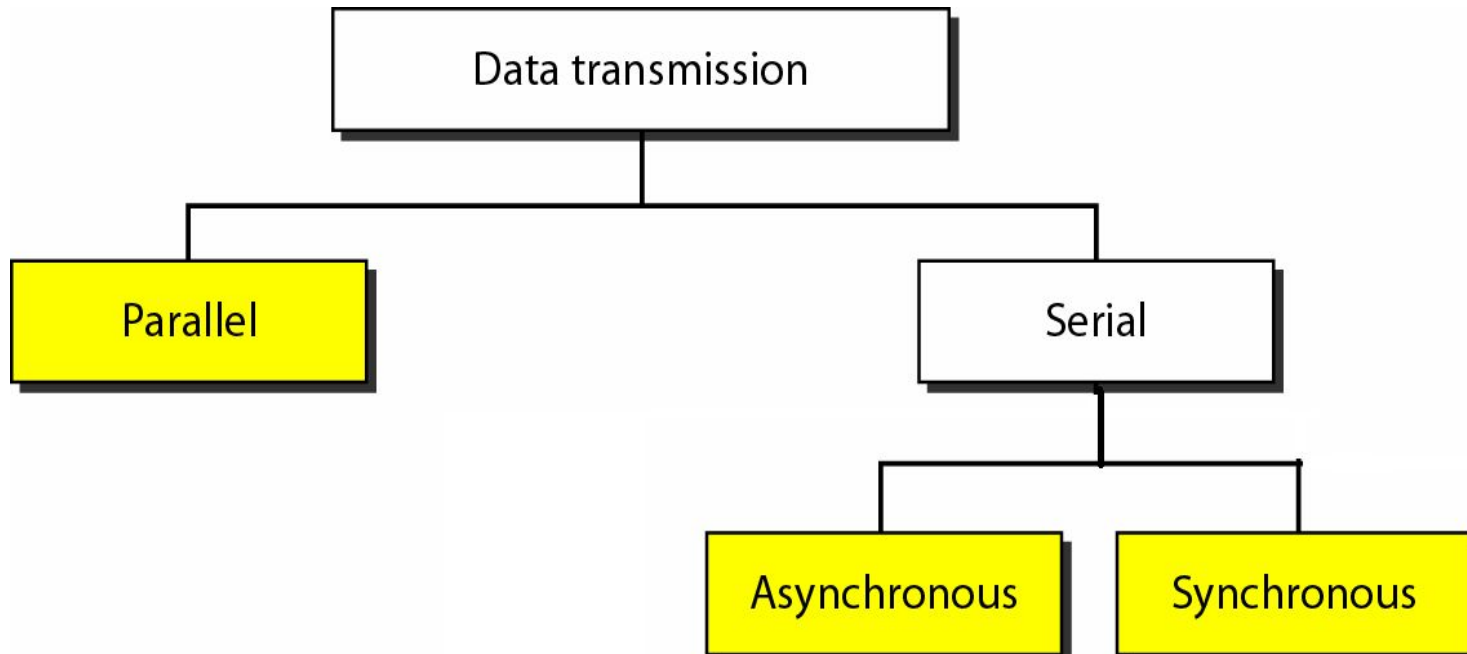
Note

According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

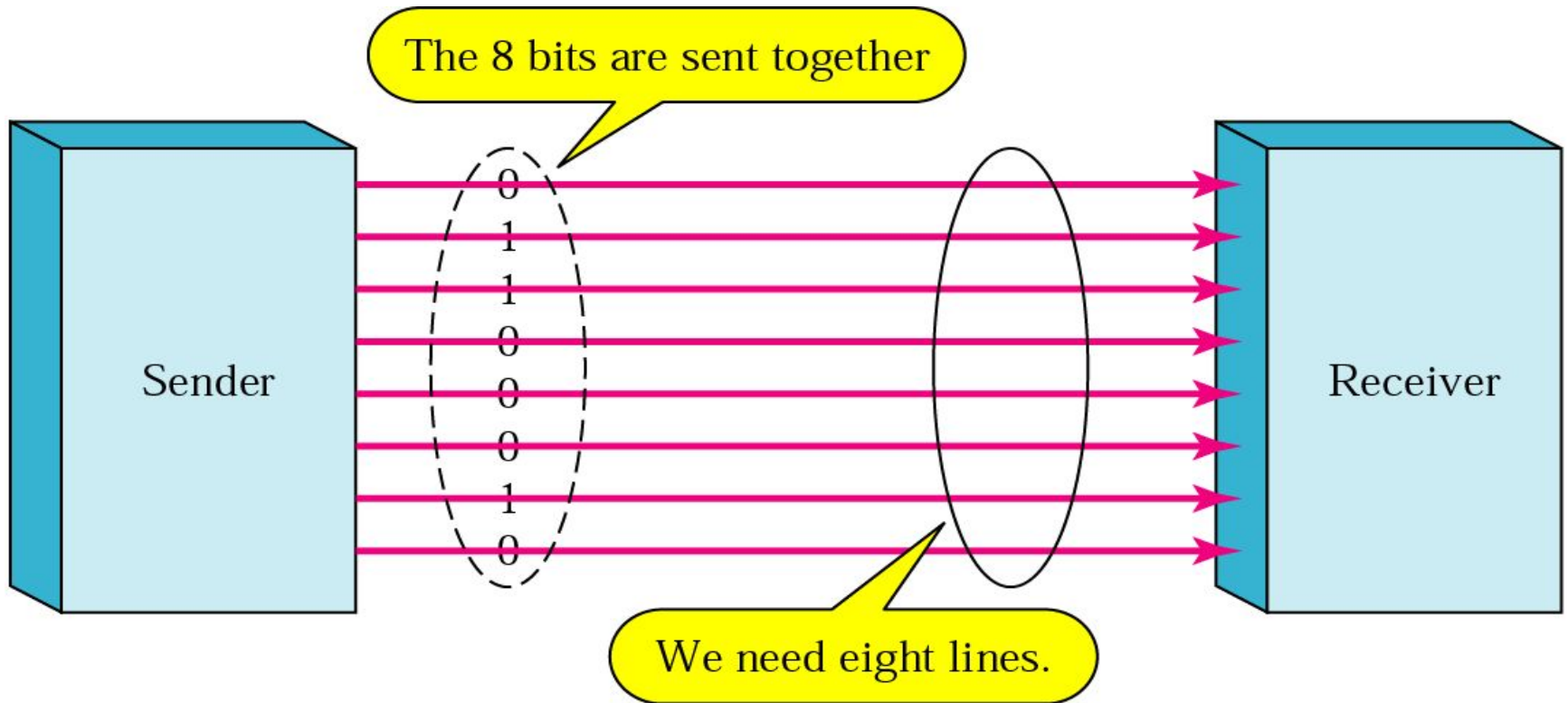
Transmission Modes

Transmission Modes

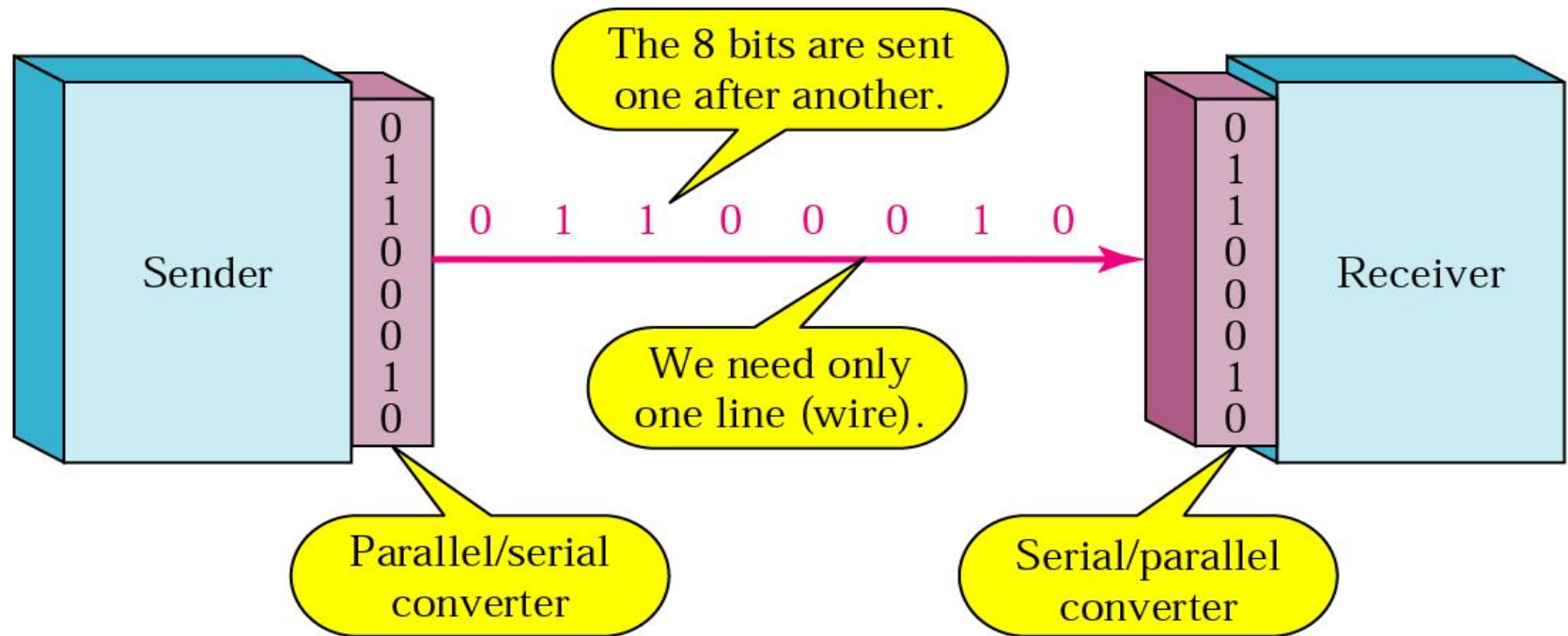
- The transmission of binary data across a link can be accomplished in either parallel or serial mode. In parallel mode, multiple bits are sent with each clock tick. In serial mode, 1 bit is sent with each clock tick. While there is only one way to send parallel data, there are two subclasses of serial transmission: asynchronous, synchronous.



Parallel Transmission



Serial Transmission



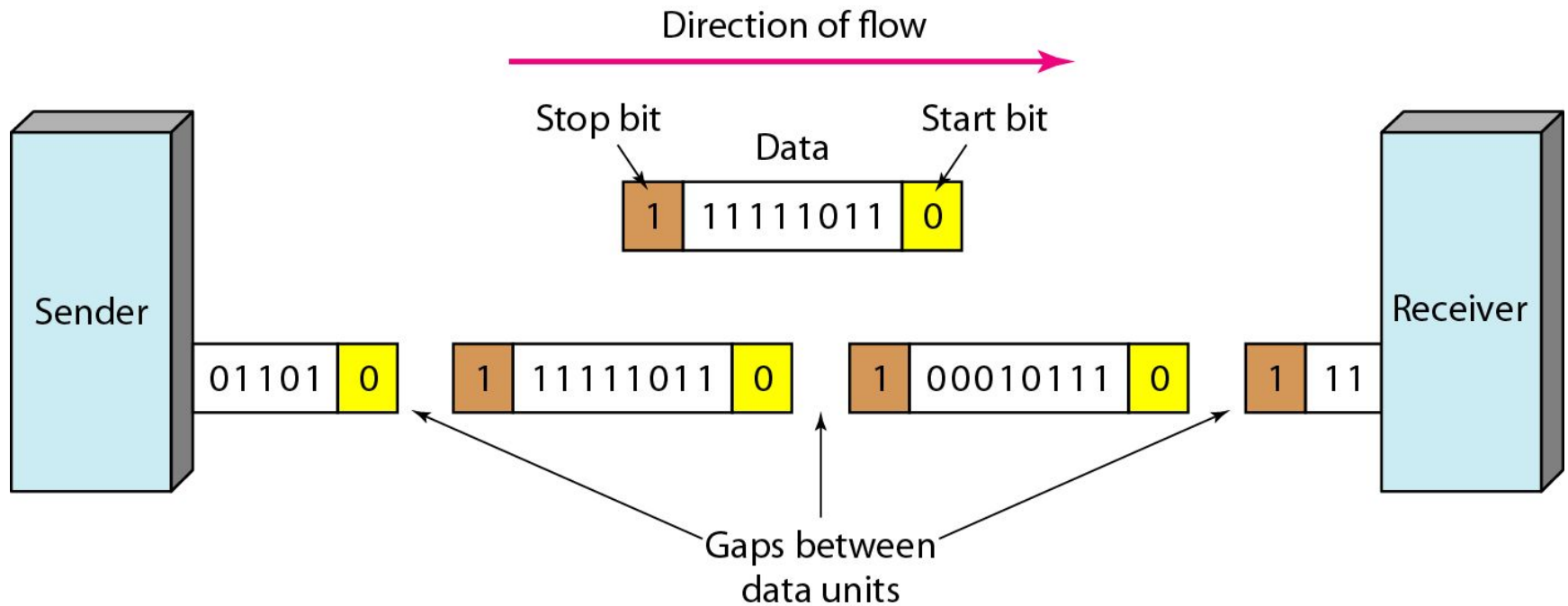
Note

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Note

**Asynchronous here means
“asynchronous at the byte level,”
but the bits are still synchronized;
their durations are the same.**

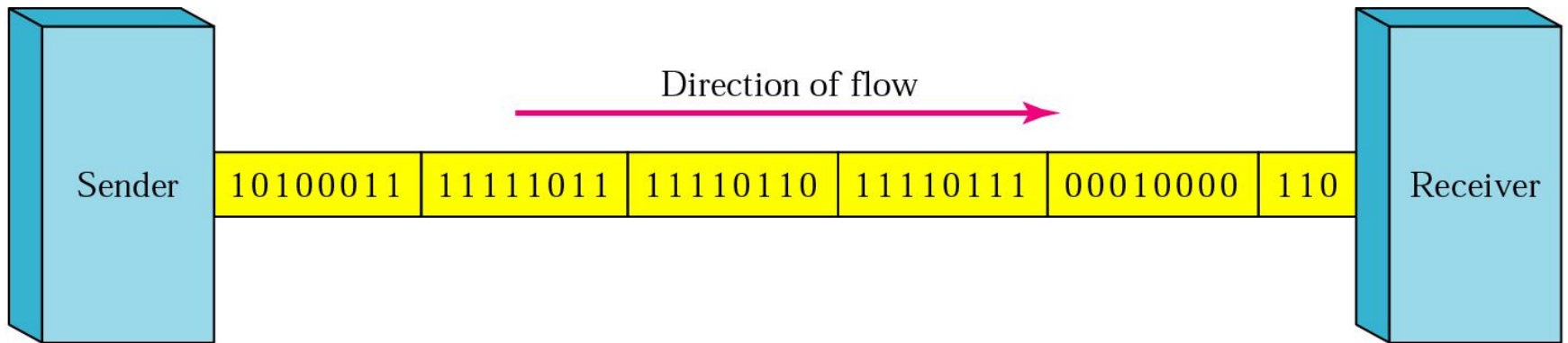
Asynchronous Transmission



Note

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

Synchronous Transmission



Readings

- Chapter 4 (B.A Forouzan)
 - Section 4.1, 4.2, 4.3

